



CSE472: HCI – Human Computer Interaction

Lecture – 6: Qualitative Analysis
Spring 2026

"Draw a cute picture for my HCI class to understand qualitative analysis or thematic analysis to my students". [Gemini AI]



Inspiring Excellence

Qualitative Analysis

- So, you have collected data from all the qualitative research you have been doing.
- Contextual inquiries, interviews, surveys, observations...
- **NOW WHAT?**
- Data analysis
 - An attempt by the researcher to summarize the data.
- Data Interpretation
 - An attempt to derive meaning from the data

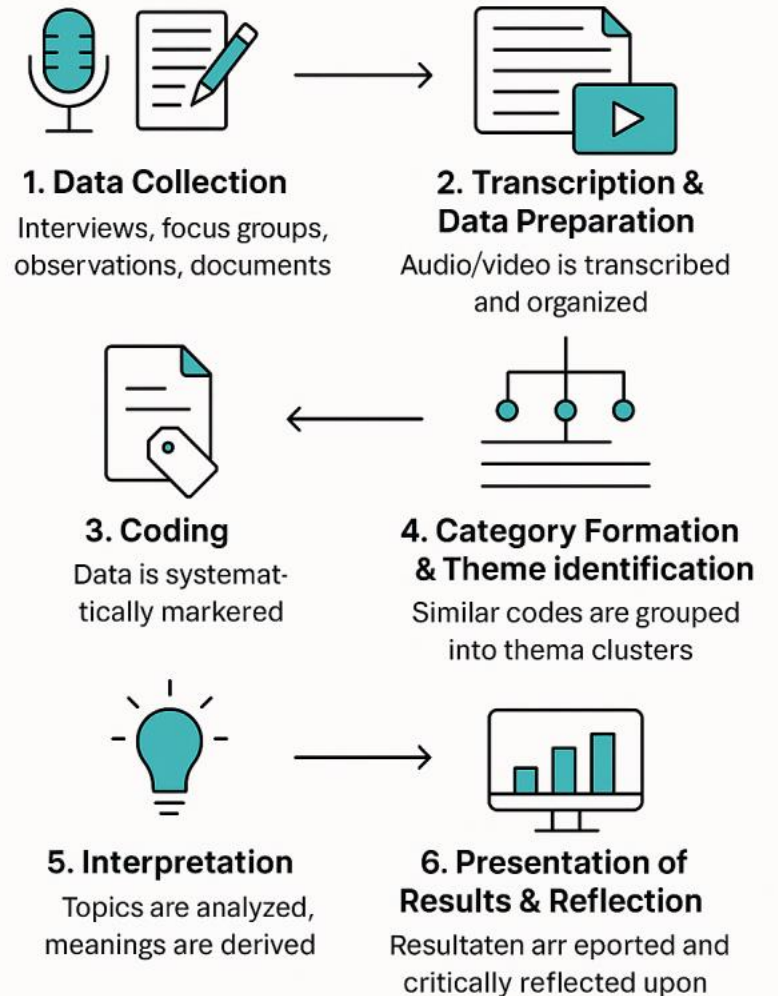


Qualitative Analysis Phases

What is Qualitative Research?

- Explores and understands human behavior and experiences
- Uses descriptive, non numeric data
- Emphasizes depth over measurement
- Flexible and open ended approach
- Focus: uncover the stories behind the data

Process of Qualitative Analysis in Phases



Inductive Content Analysis/Thematic Analysis

- There are many different ways of analyzing the data collected from qualitative methods.
- The one we'll be looking at is **inductive analysis**. Also called **thematic analysis**.
 - Deductive research → theory-testing; you start with an existing theory or hypothesis and test it with data.
 - Inductive research → theory-generating; you *start with data* and let the themes emerge from it, then build a theory.
- With inductive analysis, you generate **themes** and use them to **create theories/narratives** and **draw conclusions**

Example

- Suppose you're studying how farmers in Rangpur use mobile internet.
- Deductive: "I believe they face literacy barriers – let me confirm this."
- Inductive: "Let me talk to them with an open mind and see what themes come out." The inductive approach might uncover something surprising – maybe connectivity cost is a bigger barrier than literacy!
- **Bottom Line:** Thematic analysis lets the *data speak*, not the researcher's preconceived ideas.

How do you come up with themes?

- **Qualitative coding** is the process of going through the data and coming up with categories and meanings (themes, ideas, etc.)
- **Qualitative coding lets you make sense of and analyze your data. How?**
- **Reduces data:** 50 pages of interviews become 30-40 codes – much more manageable!
- **Allows comparison:** You can retrieve all instances of a particular code and compare them across participants.
- **Reveals patterns:** Once all your codes are visible, patterns naturally start to appear.
- **Generates theory:** Eventually, these patterns can become a broader theory about user behavior.

Sources of codes

- A priori codes i.e., Come up codes with before you analyze
 - Previous research
 - Previous theory
 - Research question
 - Your intuition of the data or setting
- Grounded codes
 - "In vivo": Let codes emerge from the data
 - Suspend your ideas about the phenomenon and let your data determine your thinking

Code Names

- Codes are given **meaningful names** that are applied to all instances of similar content.
 - Strings of text may contain *more than one code*.
 - When new content is discovered, a new code is created to apply to it and other similar content.
- As you do your analysis:
 - Codes *may evolve*
 - The number of codes may grow as more topics or themes become apparent.
- Therefore, generate and maintain *a list of codes (i.e., codebook)* to help to identify the content contained in the codes and the data set.

Keep in mind: Credibility

- Is the data based on a participant's own observation, or hearsay?
- Is there corroboration by other participants?
- In what circumstances was an observation made or reported?
- How reliable are the participants providing the data?
- What motivations might have influenced a participant's report?
- What biases might have influenced how an observation was made or reported?

Remember: Interpretation also depends on the perspective of the researcher(s)

Coding

- Codes may be based on:
 - Actions, behaviors – What users *do*
 - Themes, topics – Recurring subjects that come up
 - Ideas, concepts – Abstract thoughts or beliefs
 - Terms, phrases – Recurring language patterns
 - Keywords – Single important words
- *Code only relevant data (Not all data must be coded)*

Types of Codes Examples

- Behaviors, specific acts
 - Specific events, stories (short or once)
 - Activities (common or longer duration)
 - Strategies, practice, or tactics
 - States – general conditions/feeling
 - Relationships or interaction
 - Conditions or constraints
 - Consequences
- *Being dumbfounded, Bragging*
 - *First time visiting Cox's bazar.*
 - *Playing cricket, eating out*
 - *Doing extra homework to get an A+*
 - *Being angry about Assignment 1*
 - *I want to beat the other team*
 - *I can't use Facebook on the subway*
 - *Getting hacked because you had a weak password*

Coding process

Follow an iterative process:

1. Read through data multiple times. Become familiar with the data through reading data and transcripts, listening to recordings, etc.
2. Categorize and code pieces of data
3. Group the codes into themes
4. Reduce overlap and redundancy
5. Repeat, refine, repeat, refine...

Categorize and Code

Answer these four questions

- What is important in the data?
- Why is it important?
- What can be learned from it?
- So what?

Example:

“. . . especially if they're separated and having custody issues, the abuser will give a cellphone to one of the kids. And the kid is so excited, they get a cellphone, but [the abuser] uses it as a way of getting in and figuring out what's going on in the home.”

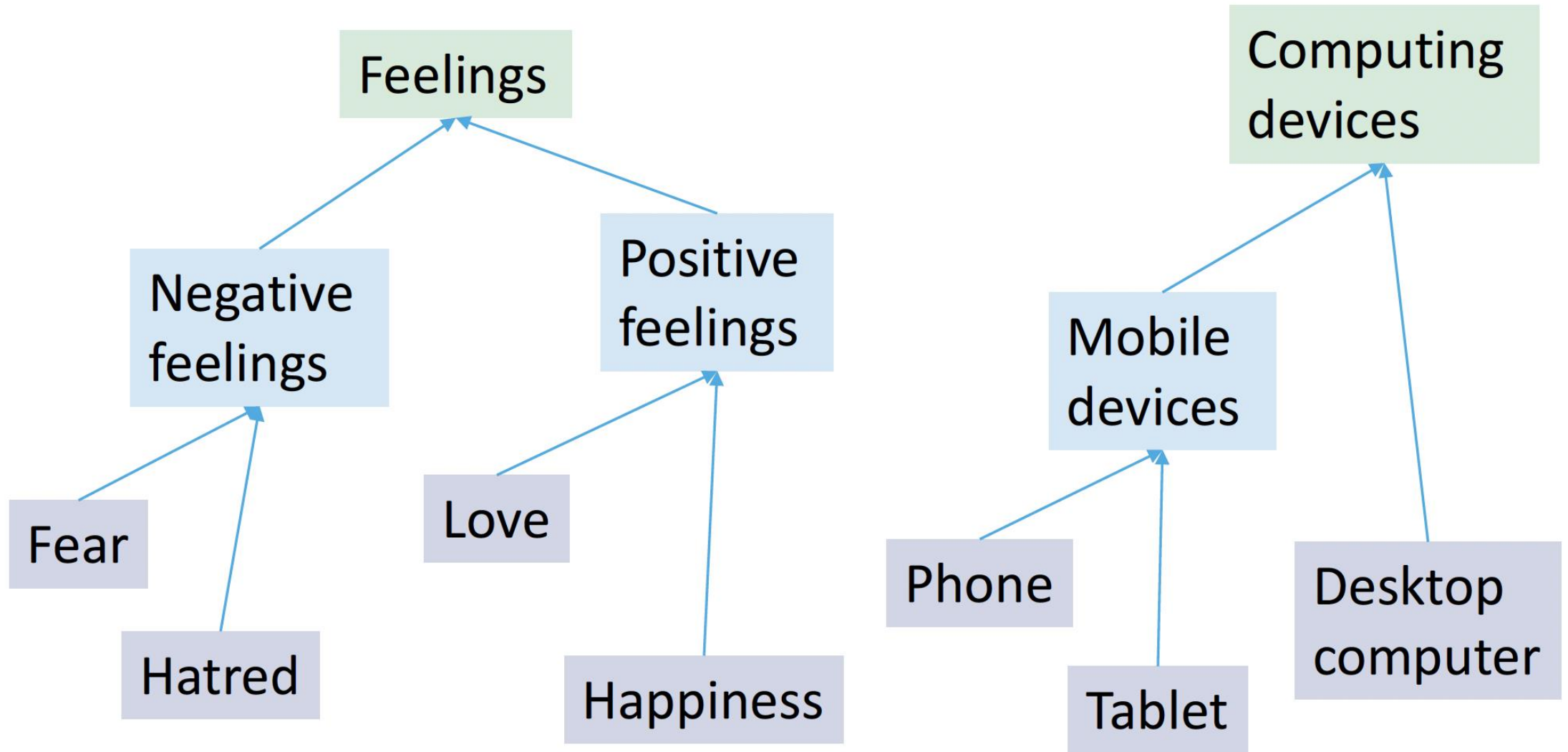
How to make coding manageable

- Make copies of the original data
 - Why?
- Read through all of the data.
 - Attach working labels to blocks of text
- Cut and paste blocks of text onto index cards.
- Group cards that have similar labels together
- Revisit piles of cards to see if clusters still hold together.
- Re-cluster into different piles. etc.

Identifying themes

- Generate broader themes by linking instances of codes with other instances/codes.
- Begin with big picture and list “themes” that emerge.
 - Events/codes that keep repeating themselves
- First round of coding: 30-40 categories
- Second round of coding: 15-20 categories
 - Remove redundancy
 - Reduce overlap
- Eventual target: 3-8 main categories/themes
 - Can have sub-categories

Identifying themes



Create a narrative

- Once you have your ~3-8 main themes, the themes are formed into a narrative about the data.
- Create a story that best represents your data
 - “Our participants expressed mixed feelings about deleting their Facebook accounts. For example....”
 - You can’t include every detail!
- For product design, create “user stories” that describe the key concerns/actions/feelings for each main category of user.
 - Often several important categories of users.
- We will learn more about user stories, personas, user journeys, and how to translate them into designs in upcoming classes.

Summary of Qualitative Analysis

1. Initial read through of the data (many pages of text)
2. Identify specific segments of information (many segments of text)
3. Label the segments to create categories (30-40 codes)
4. Reduce overlap and redundancy among categories (15-20 codes)
5. Create a model incorporating the most important categories (3-8 categories, with sub-categories)
6. Generate a narrative/user stories from your most important codes/themes

Thematic Analysis: A thorough Example

Research Topic: Perceptions and Challenges of Using Mobile Financial Services (MFS) like bKash or Nagad among Small Business Owners in Dhaka

Research Question: "What are the primary facilitators and barriers for small business owners in Dhaka when using MFS for their daily transactions?"

One of the interview questions: "Can you tell me about your experience using mobile banking for your business? What works well and what are the challenges?"

A thorough Example: Responses

P1 response: *"Using bKash has been a lifesaver, honestly. Customers can pay me instantly, even if they don't have cash. I don't have to worry about giving exact change. But sometimes, the network is so bad, especially in the evening. The transaction fails, and the customer gets frustrated. Also, the cash-out fees are a bit high. When you have many small transactions, it adds up and cuts into my profit. And I worry about scams... you hear stories of people calling and tricking you to get your PIN."*

P2 response: *"For me, Nagad is very good. Most students use it. It is fast, and I get the money right away. It's much safer than keeping a lot of cash in my drawer, which could get stolen. My main problem is that many of my older suppliers don't use it. They only want cash. So I still have to go to an agent to cash out regularly, which takes time out of my day. Sometimes the app is also confusing... too many menus and options. I just want to receive and send money, that's it."*

A thorough Example: Responses

P3 response: *"My entire business runs on mobile payments! It's essential. I can get orders from all over Dhaka and they pay me through bKash before I deliver. It gives me security that I won't be cheated. But I am always scared of making a mistake, like sending money to the wrong number. It has happened once, and getting it back was a nightmare. Also, the transaction limits are a problem. On a good day, I hit the limit and then I can't accept any more payments until the next day, which means I lose business."*

Let's code these responses: A code is a short phrase that captures the essence of a segment of the text by categorizing and labeling pieces of data. Now, let's break down these responses and assign initial codes.

A thorough Example: Initial Coding (1)

Participant	Raw Data Snippet (translated from Bengali)	Initial Codes
P1	<i>"Customers can pay me instantly, even if they don't have cash."</i>	Convenience of digital payment
	<i>"I don't have to worry about giving exact change."</i>	Eliminates change issue
	<i>"sometimes, the network is so bad... the transaction fails."</i>	Poor network reliability
	<i>"the cash-out fees are a bit high... it adds up."</i>	High transaction cost

A thorough Example: Initial Coding (2)

Participant	Raw Data Snippet (translated from Bengali)	Initial Codes
P2	<i>"It is fast, and I get the money right away."</i>	Speed of transaction
	<i>"It's much safer than keeping a lot of cash in my drawer."</i>	Improved physical security
	<i>"many of my older suppliers don't use it. They only want cash."</i>	Limited ecosystem adoption
	<i>"I still have to go to an agent to cash out regularly, which takes time."</i>	Time cost of cashing out
	<i>"Sometimes the app is also confusing... too many menus."</i>	Complex user interface

A thorough Example: Initial Coding (3)

Participant	Raw Data Snippet (translated from Bengali)	Initial Codes
P3	<i>"My entire business runs on mobile payments! It's essential."</i>	Business dependency on MFS
	<i>"they pay me through bKash before I deliver. It gives me security."</i>	Payment security (pre-delivery)
	<i>"I am always scared of making a mistake, like sending money to the wrong number."</i>	User error anxiety
	<i>"getting it back was a nightmare."</i>	Poor recourse for errors
	<i>"the transaction limits are a problem... I lose business."</i>	Transaction limits as barrier

Creating Themes from Codes

Code Cluster	Supporting Codes	Resulting Theme
Positive Experiences (Facilitators)	- Convenience of digital payment - Eliminates change issue - Speed of transaction - Improved physical security - Business dependency on MFS - Payment security (pre-delivery)	"Perceived Improvement in Transaction Efficiency and Security"
Negative Experiences (Barriers)	- Poor network reliability - High transaction cost - Time cost of cashing out - Transaction limits as barrier	"Operational and Financial Constraints"
Fears and Usability Issues	- Security fear: Scams - Complex user interface - User error anxiety - Poor recourse for errors - Limited ecosystem adoption	"Mistrust and Usability Hurdles"

Popular Tools for Thematic Analysis (CAQDAS)

Several popular CAQDAS (Computer-Assisted Qualitative Data Analysis Software):

1. NVivo

Best for: Large-scale academic and professional research projects.

Key Features:

- Handles diverse data types (text, audio, video, social media).
- Powerful coding and querying tools to explore relationships.
- AI-assisted auto-coding for initial theme identification.
- Advanced data visualization (charts, models, word clouds)

Popular Tools for Thematic Analysis (CAQDAS)

2. MAXQDA

Best for: Academic researchers, especially those using mixed methods.

Key Features:

- Highly intuitive four-window interface makes it easy to learn.
- Strong memoing capabilities to document the analysis process.
- Excellent for integrating qualitative data with quantitative data (e.g., surveys).
- AI Assist helps summarize coded data and suggest new themes

Popular Tools for Thematic Analysis (CAQDAS)

3. ATLAS.ti

Best For: Academic research and complex projects needing in-depth analysis.

Key Features:

- Powerful AI for coding and querying.
- Interactive network views to map complex relationships.

4. Dovetail

Best For: UX and product teams in industry settings.

Key Features:

- Real-time collaborative tagging.
- Creates a searchable "Insight Repository" for company-wide access.

Any questions?